

CHAPTER 1

Scenario

Cloud computing is one of the most important technologies used by organizations for storing data, managing applications, and improving business operations. Traditional on-premise infrastructure requires high maintenance cost, physical hardware, and limited scalability. To overcome these issues, companies are moving towards cloud platforms such as Microsoft Azure.

This project focuses on implementing a secure cloud infrastructure for a small company using Microsoft Azure services. The infrastructure includes Virtual Machines, Storage Accounts, Blob Storage, Virtual Networks, and Role-Based Access Control (RBAC). These services help in secure resource management, centralized storage, remote accessibility, and controlled user permissions.

The project demonstrates how Azure cloud services can provide a scalable, secure, and cost-effective solution for small business environments.

CHAPTER 2

Literature Review

The idea for this project was inspired by the growing adoption of cloud computing technologies in modern organizations. Many companies are migrating their traditional IT infrastructure to cloud platforms for better scalability, security, and remote accessibility.

The project concept was studied using:

- Microsoft Azure official documentation
- Cloud Computing concepts from academic materials
- Microsoft Learn platform
- Azure Portal practical implementation guides

Cloud services such as Virtual Machines, Storage Accounts, Virtual Networks, and RBAC are widely used in organizations for managing secure cloud infrastructure. Based on these concepts, this project was developed using Microsoft Azure.

CHAPTER 3

Project Description

The project implements a secure cloud infrastructure for a small company using Microsoft Azure services.

Services Used

- Azure Virtual Machine
- Azure Storage Account
- Azure Blob Storage
- Azure Virtual Network
- Azure RBAC

Workflow

1. A Resource Group was created in Azure.
2. A Virtual Machine was deployed to simulate a company server.
3. A Storage Account and Blob Container were created for file storage.
4. Files were uploaded to Blob Storage.
5. A Virtual Network was configured for secure communication.
6. RBAC was implemented to manage user permissions securely.

Objectives

- Secure cloud resource management
- Centralized file storage
- Secure networking
- Access control management
- Scalable infrastructure

CHAPTER 4

Results and Screenshots

The cloud infrastructure was successfully implemented on Microsoft Azure.

Implemented Resources

- Resource Group created successfully
- Virtual Machine deployed and running
- Storage Account configured successfully
- Blob Container created
- Files uploaded to Blob Storage
- Virtual Network configured
- RBAC role assignment completed

IMPLEMENTATION STEPS

Step 1 — Create Resource Group

Description

A Resource Group named `CompanyInfraRG` was created in Microsoft Azure to organize and manage all cloud resources used in the project.

Output

Resource Group was created successfully.

Figure 1:

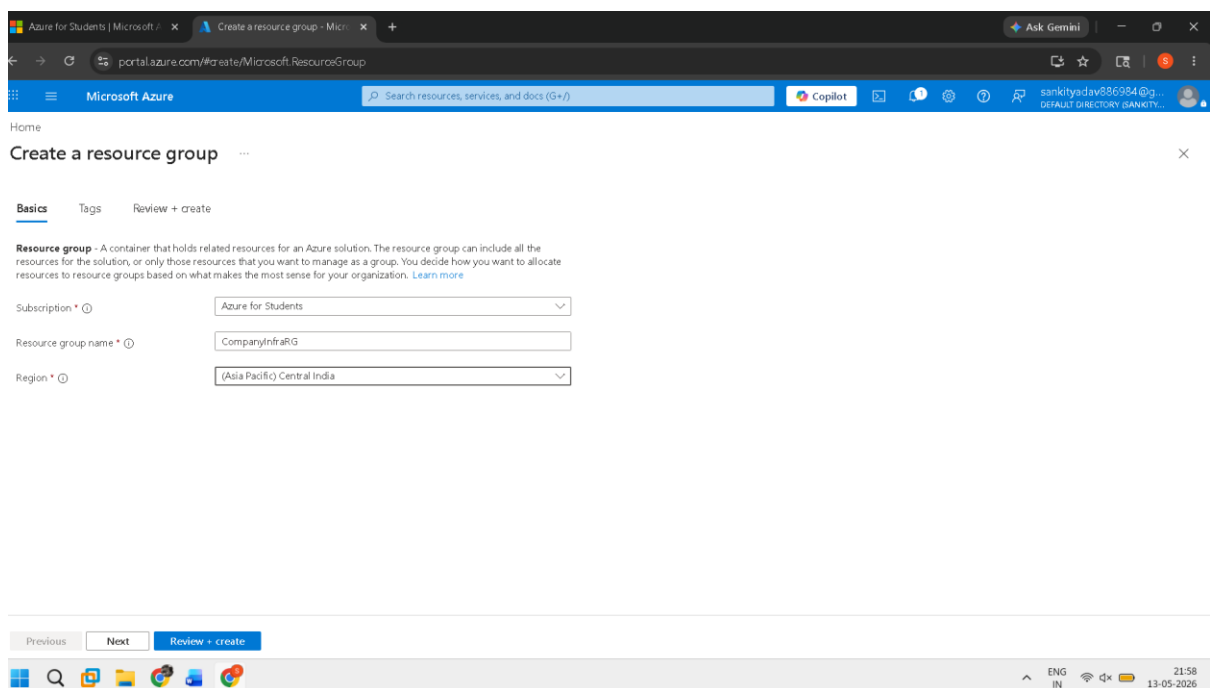


Figure 2:

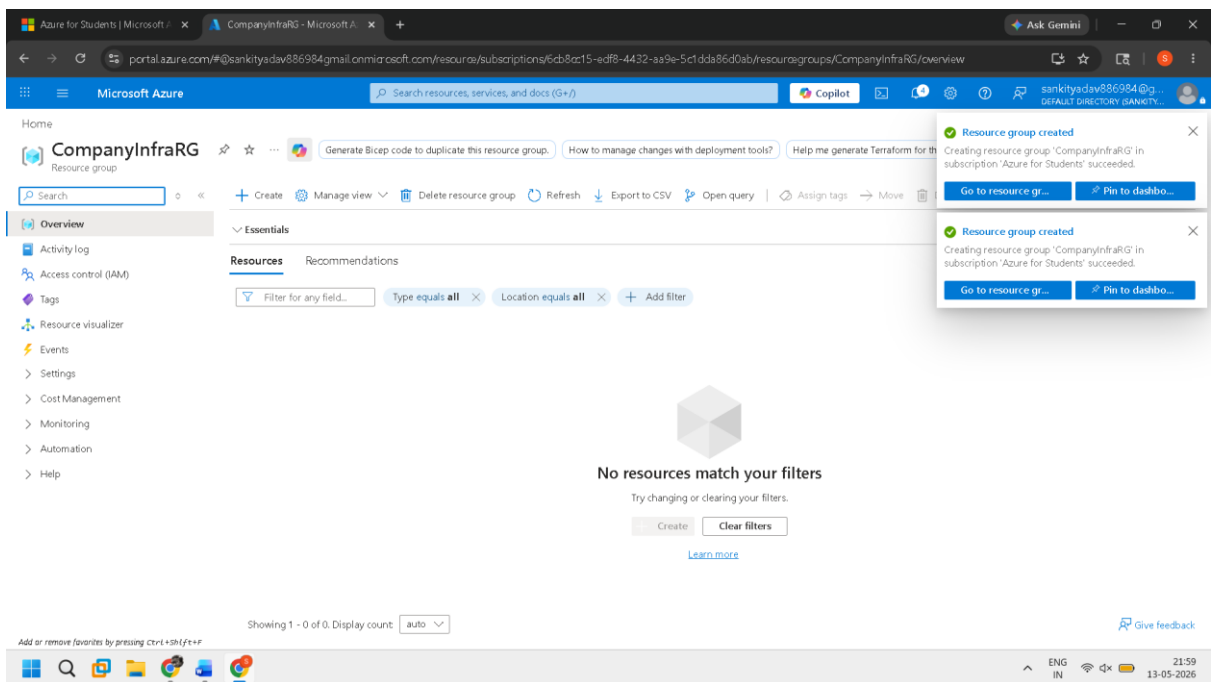
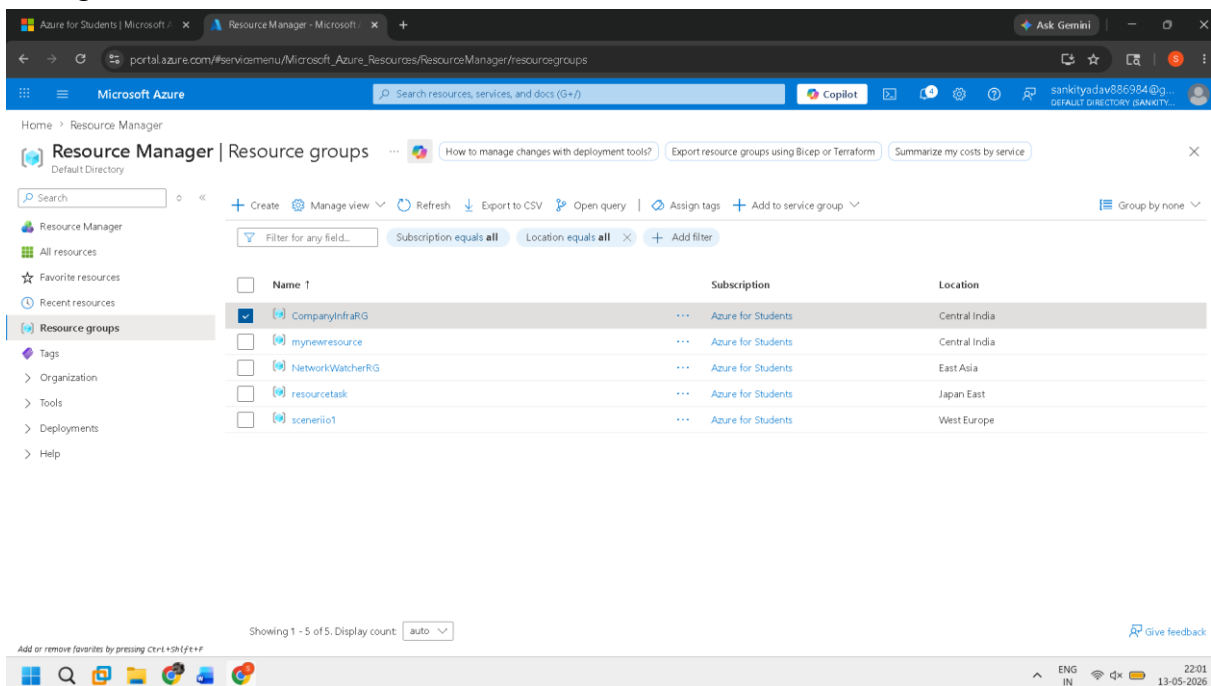


Figure 3:



STEP 2 — Create Virtual Machine

Description

A Virtual Machine named `CompanyServerVM` was created in Microsoft Azure to simulate a company server for cloud infrastructure management.

Output

Virtual Machine was deployed and running successfully.

Figure 4:

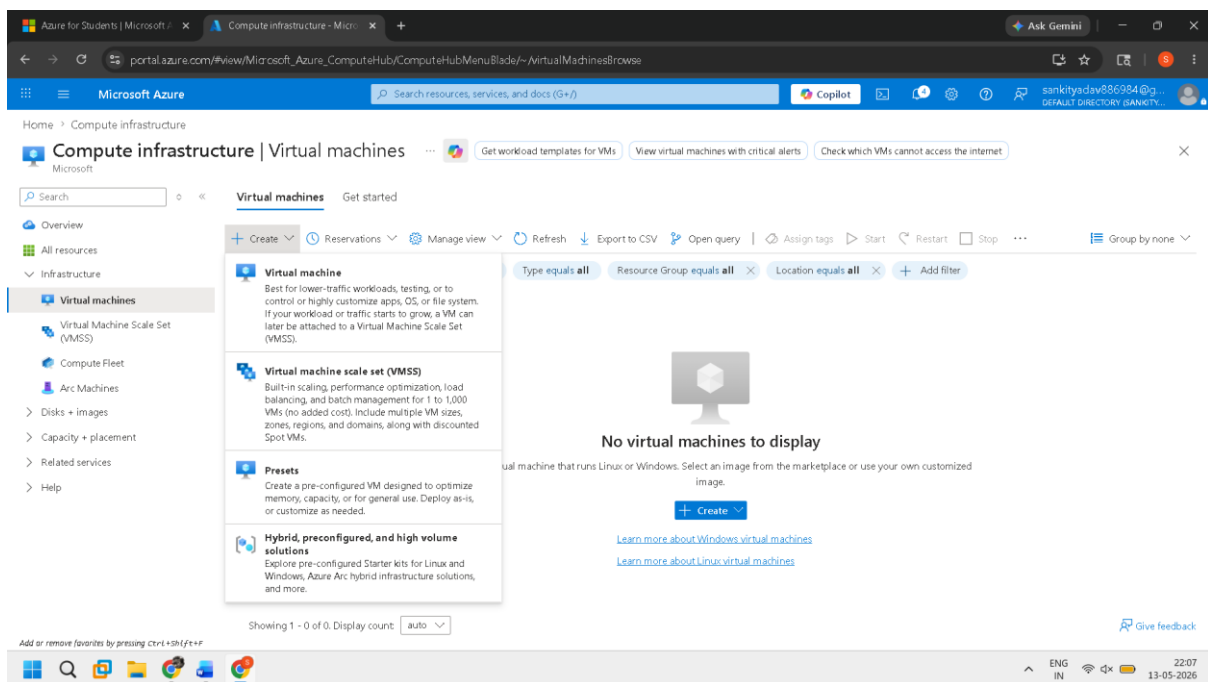


Figure 5:

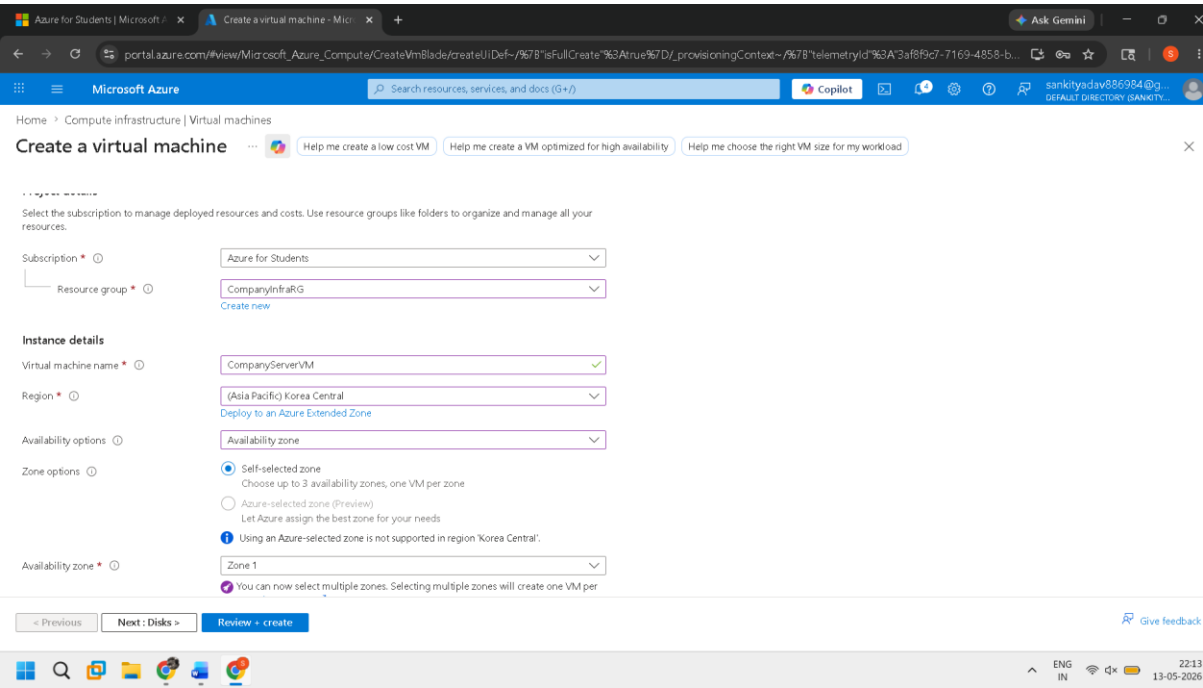


Figure 6:

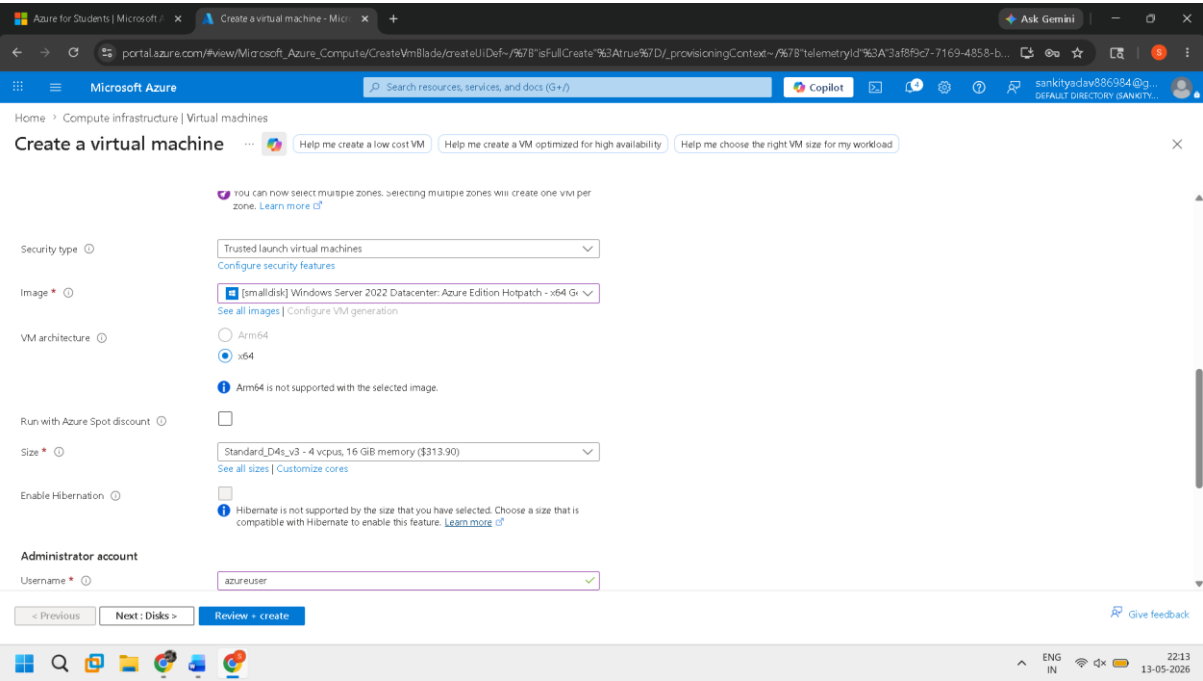
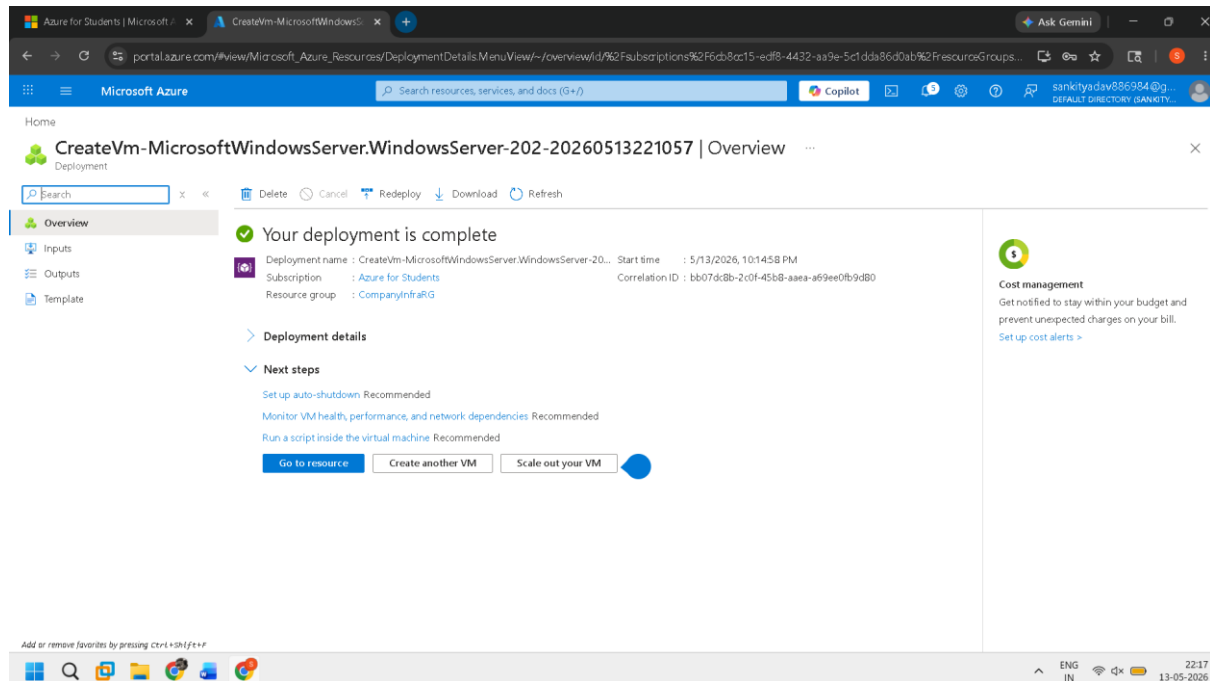


Figure 7:



STEP 3 — Create Storage Account

Description

A Storage Account was created to store company files and cloud data securely.

Output

Storage Account created successfully.

Figure 8:

Azure for Students | Microsoft / x Create a storage account - Mic x + Ask Gemini

portal.azure.com/#view/Microsoft_Azure_Storage/CreateStorageAccount_ReadView/provisioningContext~/~/678?initialValues%3A%7B%22subscriptionId%3A%5B%226db8ac15-edf8-4432-...

Microsoft Azure Search resources, services, and docs (G+)

Home > Storage center | Blob Storage

Create a storage account

Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below.
[Learn more about Azure storage accounts](#)

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription * Azure for Students

Resource group * CompanyInfraRG
[Create new](#)

Instance details

Storage account name * companystorage12345

Region * (Asia Pacific) East Asia
[Deploy to an Azure Extended Zone](#)

Preferred storage type Azure Blob Storage or Azure Data Lake Storage Gen 2

This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. [Learn more](#)

Performance * Standard Recommended for most scenarios (general-purpose v2 account)

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Figure 9:

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portal.azure.com/#view/Microsoft_Azure_Storage/CreateStorageAccount_ReadView/provisioningContext~/~/678?initialValues%3A%7B%22subscriptionId%3A%5B%226db8ac15-edf8-4432-...

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Home > Storage center | Blob Storage

Create a storage account

Basics Advanced Networking Data protection Security Encryption Tags Review + create

[View automation template](#)

Basics

Subscription	Azure for Students
Resource group	CompanyInfraRG
Location	East Asia
Storage account name	companystorage12345
Primary service	Azure Blob Storage or Azure Data Lake Storage Gen 2
Performance	Standard
Replication	Locally-redundant storage (LRS)

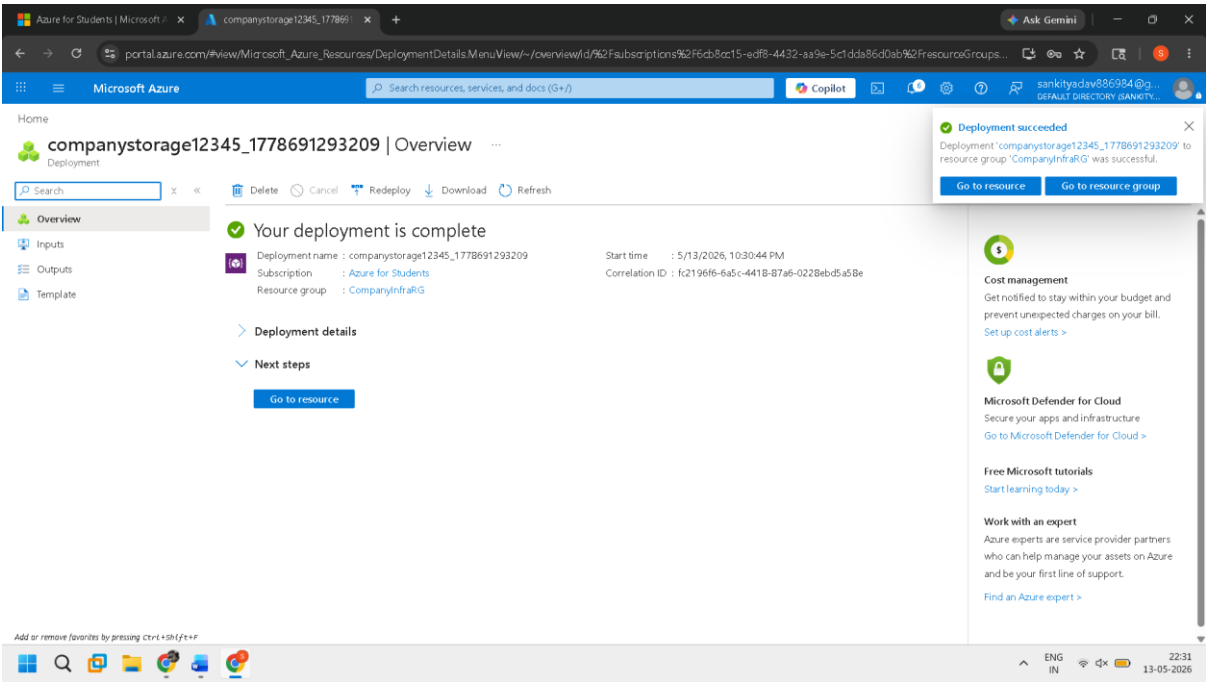
Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Managed Identity for SMB	Disabled

Previous Next Create

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Figure 101:



STEP 4 — Create Blob Container

Description

A Blob Container named `companyfiles` was created inside the Storage Account to upload and manage files.

Output

Blob Container created successfully.

Figure 11:

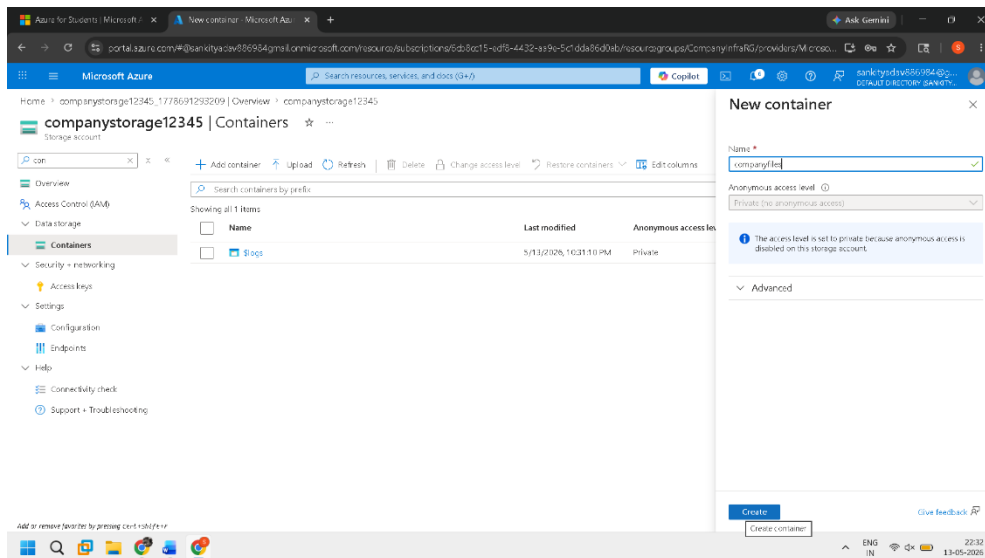
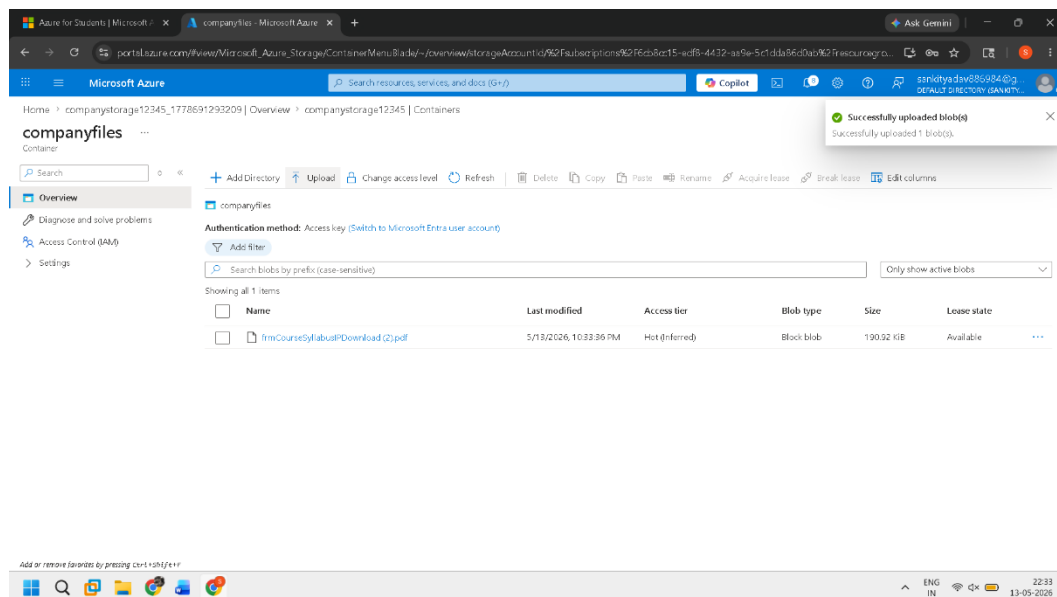


Figure 12:

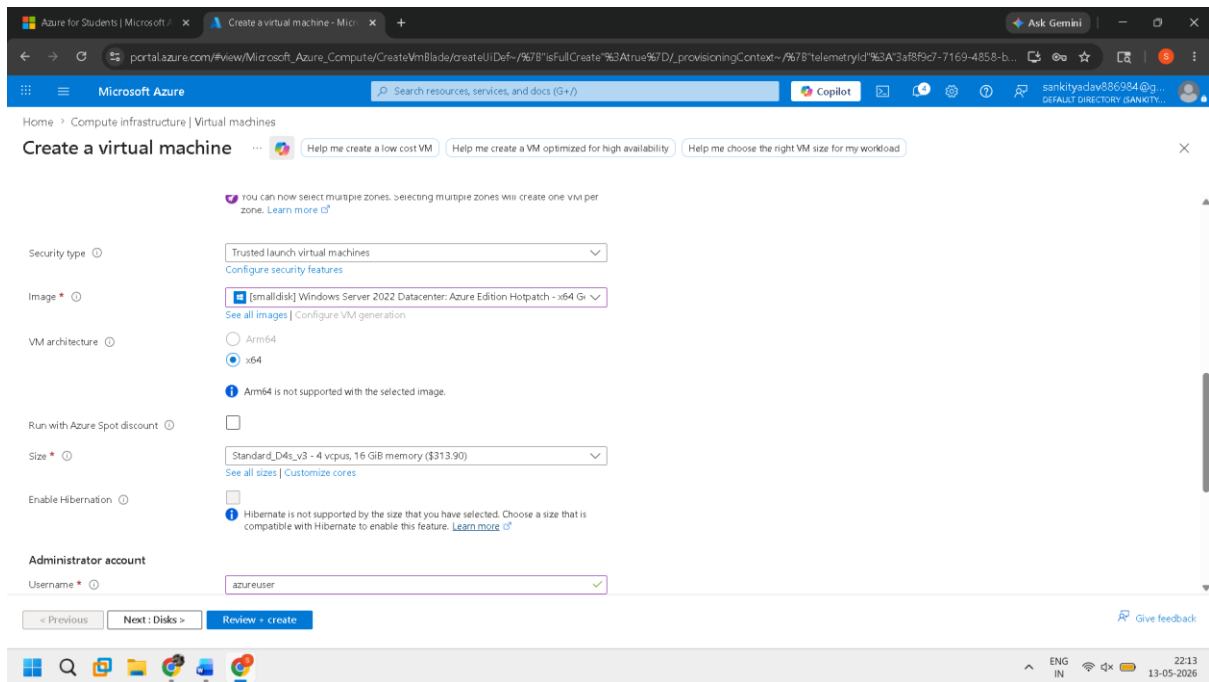


STEP 5 — Configure Virtual Network

Description

A Virtual Network was configured to provide secure communication between Azure resources.

Figure 13



Output

Virtual Network configured successfully.

STEP 7 — Configure RBAC

Description

Role-Based Access Control (RBAC) was configured to manage secure user permissions.

Output

RBAC configured successfully.

Figure 14:

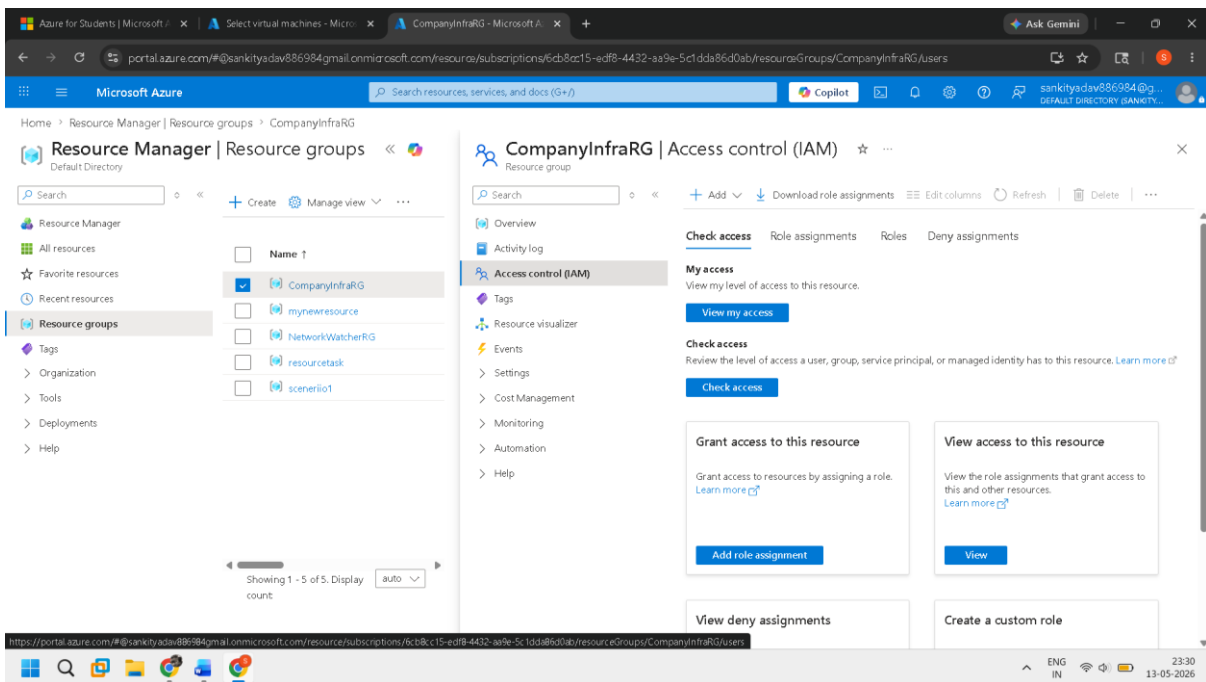


Figure 15:

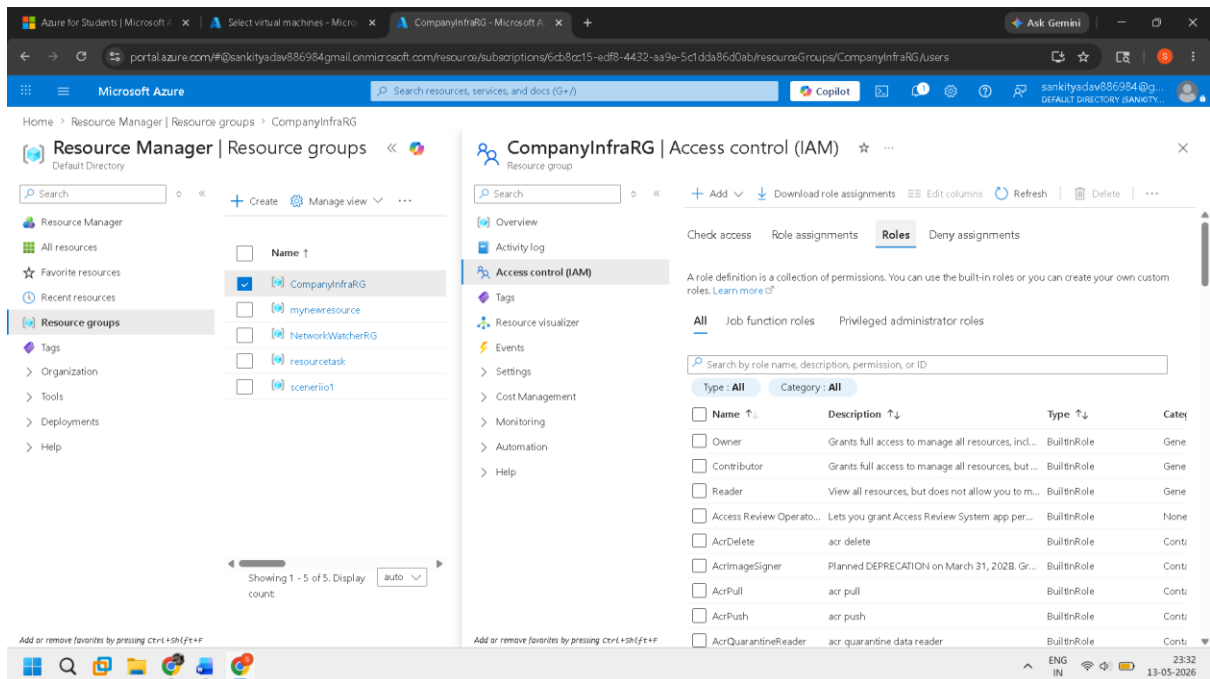
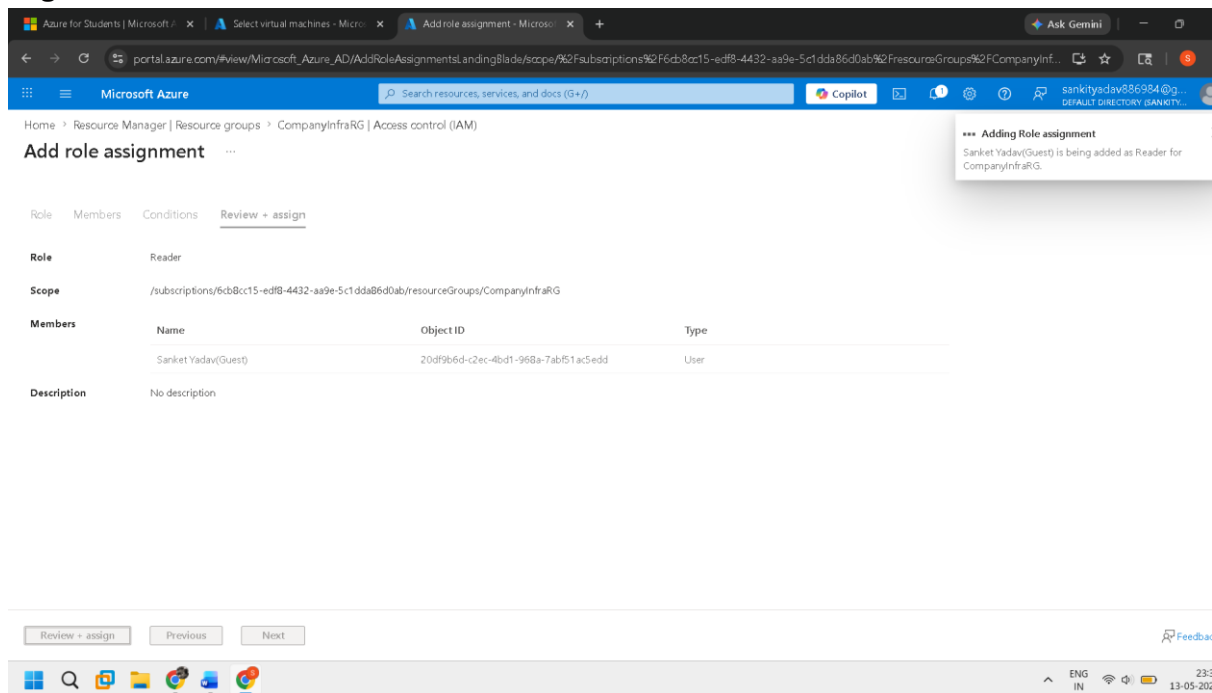


Figure 16:



Conclusion

The project “Implementation of Secure Cloud Infrastructure Using Microsoft Azure” was completed successfully using Microsoft Azure services. Different Azure resources such as Virtual Machine, Storage Account, Blob Storage, Virtual Network, and RBAC were created and configured successfully.

The project helped in understanding practical implementation of cloud computing concepts, secure cloud storage, networking, and access control management using Microsoft Azure. The implemented cloud infrastructure provides secure, scalable, and cost-effective cloud services for small organizations.